End of Topic Test – Chapter 2: Personal Finance (Test 2) — Mark Scheme

Total: 71 marks

1. Budgeting and current account

(a) B1 "OD" means overdrawn/overdraft – the account balance has gone below £0 (the customer has spent more than they had). B1 the unauthorised OD fee was charged because the worker went overdrawn without having an agreed/authorised overdraft facility in place. *[2]*

(b) M1 total income = salary = £860.00 (only credit entry). M1 total expenditure = 15.00 + 40.00 + 6.00 + 55.00 + 210.00 + 400.00 + 48.75. A1 total expenditure = £774.75. A1 both totals correctly identified and clearly labelled (income vs. expenditure). *[4]*

(c) B1 suggestion 1 (e.g. set up balance alerts / check balance before spending). B1 explanation of how it helps (avoids spending when balance is low, so avoids unauthorised OD fees). B1 suggestion 2 with explanation (e.g. build a small savings buffer / reduce discretionary spending such as online shopping, so a cushion exists before the next salary payment). *[3]*

2. Income tax bands

Personal allowance \$12,570; 20% on \$12,571–\$50,270; 40% above \$50,270.

(a) M1 taxable income = 45,200 − 12,570 = 32,630. M1 tax = 0.20 × 32,630. A1 tax = \$6,526. *[3]*

(b) M1 tax in basic band = 0.20 × (50,270 − 12,570) = 0.20 × 37,700 = \$7,540. M1 tax in higher band = 0.40 × (76,000 − 50,270) = 0.40 × 25,730 = \$10,292. M1 total = 7,540 + 10,292. A1 total tax = \$17,832. *[4]*

(c) M1 average rate = 17,832 ÷ 76,000 × 100. A1 = 23.5% (1 d.p.). *[2]*

3. Payslip and deductions

(a) B1 gross pay is total earnings before any deductions; net pay is the amount actually received after tax, NI, and other deductions are subtracted. *[2]*

(b) M1 total deductions = 350.60 + 195.84 + 140.00 = \$686.44. M1 net pay = 2800.00 − 686.44. A1 net pay = \$2,113.56. A1 method and answer consistent with the payslip. *[4]*

(c) M1 income above threshold = 2800 − 1048 = \$1,752. M1 12% × 1,752 = \$210.24. A1 the calculated deduction (\$210.24) does *not* match the \$195.84 shown on the payslip – candidates should identify this inconsistency; award the mark for a correct, clearly stated comparison (full credit for correct method and arithmetic even though the figures do not reconcile – this is a genuine discrepancy in the data given). *[3]*

4. Exchange rates

(a) M1 amount after flat fee = 600 − 5 = £595. A1 euros = 595 × 1.14 = €678.30. *[2]*

(b) M1 commission = $0.015 \times 600 =$ £9.00. M1 amount converted = $600 - 9 =$ £591.00. A1 euros = $591 \times 1.11 =$ €656.01. *[3]*

(c) M1 correct comparison of €678.30 (Bureau A) with €656.01 (Bank B). A1 difference = 678.30 − 656.01 = €22.29. B1 conclusion clearly stated: Bureau A gives better value, by €22.29. *[3]*

5. Comparing loans using APR

Amount financed (after \$150 deposit) = \$1,050; two payments of \$600 at the end of years 1 and 2.

(a) B1 amount to be financed identified as \$1,050. M1 each payment discounted: $\dfrac{600}{1+i}, \dfrac{600}{(1+i)^2}$. A1 equation: $1050 = \dfrac{600}{1+i} + \dfrac{600}{(1+i)^2}$. *[3]*

(b) M1 trial value substituted (e.g. $i = 0.09 \Rightarrow$ PV $\approx$ \$1,055, too high). M1 second, closer trial (e.g. $i = 0.095 \Rightarrow$ PV $\approx$ \$1,048). A1 $i \approx 9.5\%$ (accept 9.0%–9.5%). *[3]*

(c) B1 the description is misleading – although no interest is charged by name, the customer repays $150+2\times600 =$ \$1,350 for a sofa worth \$1,200 cash once financed, which is equivalent to an effective interest rate. B1 supported using part (b): this equates to an APR of about 9.5%, so the deal is not truly "interest-free". B1 clear overall conclusion. *[3]*

6. Comparing savings using AER

Nominal annual rate 3.6%, compounded monthly.

(a) M1 monthly rate = $3.6\% \div 12$. A1 = 0.3% per month. B1 expressed as a decimal, 0.003. *[3]*

(b) M1 method: each month's interest = starting value $\times 0.003$, rounded to the nearest penny, added to give the next starting value.

Month	Starting (\$)	Interest (\$)	Final (\$)
1	1500.00	4.50	1504.50
2	1504.50	4.51	1509.01
3	1509.01	4.53	1513.54

A1 Month 1 and Month 2 rows correct. A1 Month 3 row correct. *[3]*

(c) M1 substitute $i = 0.036$, $n = 12$ into AER = $(1 + i/n)^n - 1$. M1 $(1.003)^{12} \approx 1.03660$. A1 AER $\approx 3.66\%$ (2 d.p.). *[3]*

7. Depreciation

(a) M1 substitute into $V = 28{,}000 \times (0.85)^4$. M1 $(0.85)^4 \approx 0.52201$. A1 value $\approx$ \$14,616. *[3]*

(b) M1 total depreciation = $4 \times 5{,}200 =$ \$20,800. A1 value = $28{,}000 - 20{,}800 =$ \$7,200. B1 correct method (subtracting a fixed amount each year) clearly shown. *[3]*

(c) B1 correct comparison: reducing-balance value (\$14,616) is considerably higher than the straight-line value (\$7,200) after 4 years. B1 reducing-balance depreciation is generally more realistic for vehicles, since real resale values fall quickly at first and then more slowly, whereas straight-line depreciation falls by the same amount every year. B1 straight-line depreciation would eventually predict an unrealistic negative value if continued, which reducing-balance depreciation never does. *[3]*

8. Inflation and real value

(a) M1 expression: $45{,}000 \times 1.028^5$. M1 $1.028^5 \approx 1.14806$. A1 value $\approx$ \$51,663. *[3]*

(b) B1 the car's nominal value grows at the same 2.8% rate as general prices (inflation). B1 so the growth in money terms exactly cancels out the fall in the purchasing power of money. B1 conclusion: the real (inflation-adjusted) value of the car stays the same – it buys the same amount of goods/services as before, even though its price in \$ has risen. *[3]*

(c) M1 real growth factor $= 1.14806 \div 1.152$. M1 ≈ 0.99658. A1 real percentage change $\approx -0.3\%$ (a small real *decrease*) over the 5 years. *[3]*